

Spermidine in the aging brain: mechanisms, preclinical evidence, and clinical perspectives

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Abstract

Spermidine, a naturally occurring polyamine, has emerged as a candidate modulator of brain aging because it regulates autophagy, mitochondrial function, oxidative balance, and neuroinflammatory signaling. This review summarizes current knowledge on spermidine biology, including its synthesis, catabolism, transport, and cell-specific handling in neurons and glia, with emphasis on ATP13A2/ATP13A4-dependent trafficking and EP300-linked autophagy. Preclinical studies consistently show that spermidine enhances proteostasis, reduces soluble amyloid-beta, improves microglial phagocytic activity, preserves mitochondrial fitness, and attenuates inflammatory responses in models of aging and neurodegeneration. Human evidence is encouraging but still limited: observational studies associate higher dietary spermidine intake with better cognitive outcomes, whereas interventional trials suggest possible benefits in subjective cognitive decline and mild dementia, with variable dose-dependent effects. Overall, spermidine is a biologically plausible nutraceutical for healthy brain aging, but larger, well-designed clinical trials are needed to define efficacy, bioavailability, safety, and optimal therapeutic use across prodromal, early-stage, and disease-specific patient populations.

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